

and deployment of AI models and large language models (LLMs) so that these can be accessed on multiple interfaces, including web-based and smartphone applications. Two key platforms are discussed here.

Hugging Face Spaces with Gradio

URL: <https://huggingface.co/> and <https://www.gradio.app/>

Machine learning or deep learning models can be deployed on Hugging

Face Spaces with the integration of Gradio. The latter integrates the graphical user interface (GUI) to present the model as a web application.

A specific directory structure is followed to deploy a model on Hugging Face Spaces. Here is an example.

To deploy a classical retrieval augmented generation (RAG) application of AI, the following files are required:

- `requirements.txt`
- `app.py`

These files should be in the same folder in Hugging Face.

```
requirements.txt
transformers
gradio
faiss-cpu
sentence-transformers
torch
```

The file `requirements.txt` includes the packages that are required and used in the main `app.py` file. The Hugging Face platform installs these packages on loading and then provides the interface to execute and run the main interface as

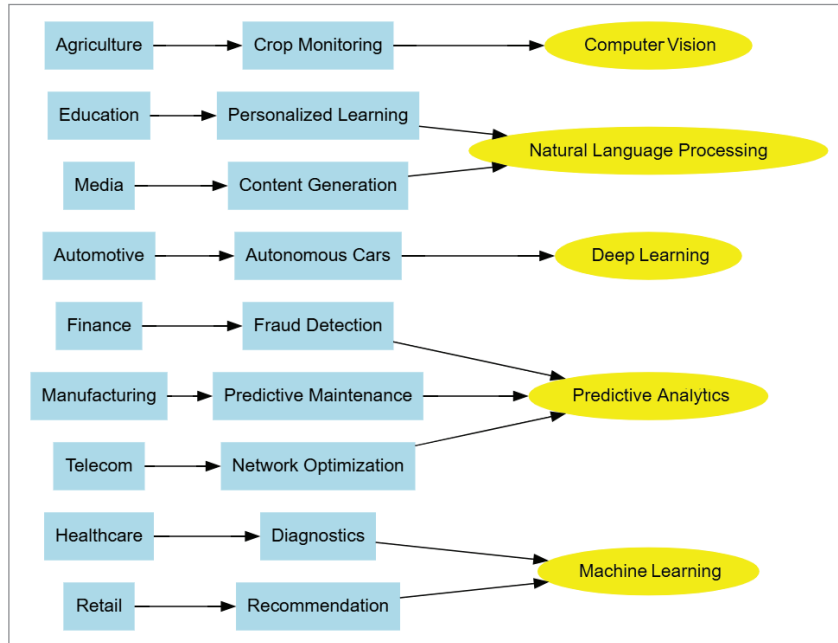


Figure 2: Key application areas and industries associated with AI

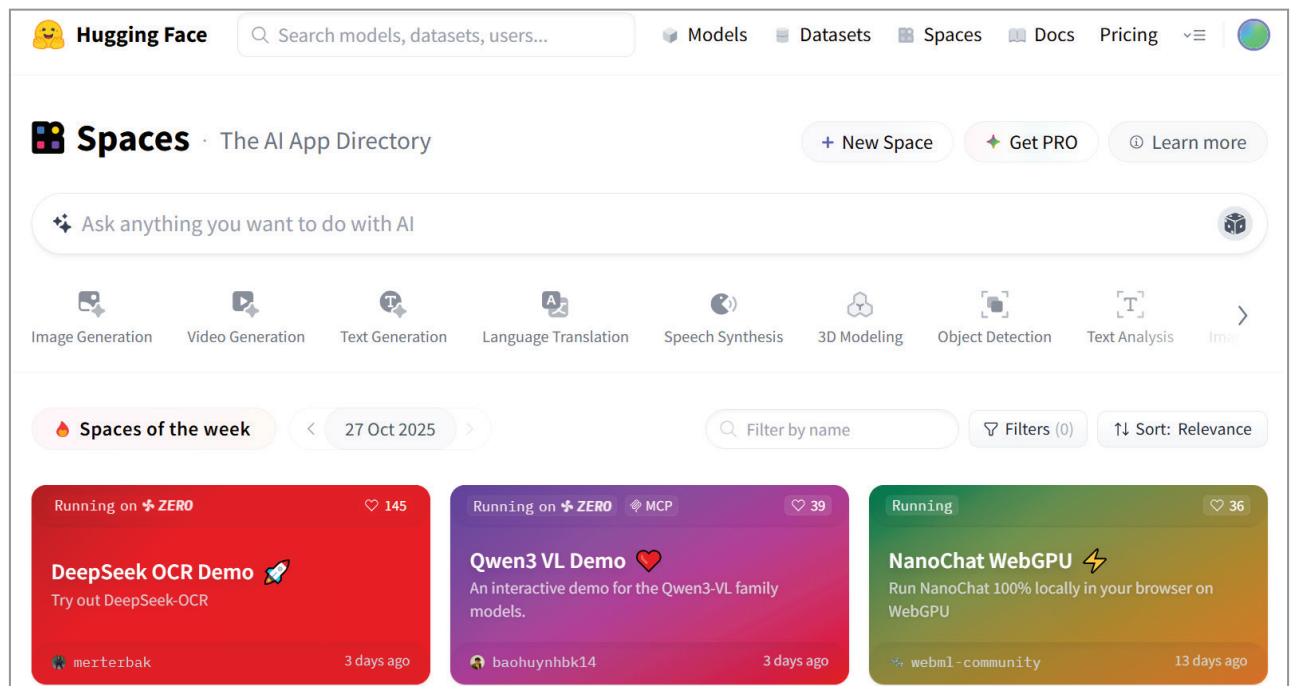


Figure 3: Hugging Face Spaces for deployment of AI models